Given a dataset $\mathcal{D} := \{(a_i, b_i) \in \mathbb{R} \times \mathbb{R} \ : \ i \in \{1, \dots, N\}\}$ generated by

```
import numpy as np
import random
random.seed(10)

# generate data set
N = 100
inputs = np.random.uniform(0, 1, size=N)
outputs = 10 * inputs + np.random.normal(size=N)

# visualize data set
import matplotlib.pyplot as plt
plt.plot(inputs, outputs, '*')
```

We consider the optimization problem

$$\min_{x \in \mathbb{R}} \frac{1}{N} \sum_{i=1}^{N} f_i(x)$$

with $f_i(x) = \left(\frac{1}{2}(a_i x - b_i)^2\right)$ for $i \in \{1, \dots, N\}$.

Exercise 13.1

The mini-batch update rule of SGD is given by

$$x^{k+1} = x^k - t \frac{1}{|B_k|} \sum_{i \in B_k} \nabla f_i(x^k),$$

where $B_k \subset \{1, \dots, N\}$ denotes the uniformly sampled batch at iteration k, $|B_k| = B$ is fixed for all iterations and $t > 0$ is a constant stepsize. We recover Stochastic Gradient Descent (SGD) with batch size $B = 1$, and full Gradient Descent (GD) with batch size $B = N$ and mini-batch SGD with $1 < B < N$.

(a) Implement GD, SGD and mini-batch SGD with different choices of B using the same stepsize t and the same number of iterations I.

(b) Plot the objective values $f(x^k) - f^\star$ versus the iteration count and the empirical variance $\mathrm{Var}(g_k \mid x^k)$, where

$$g_k = \frac{1}{|B_k|} \sum_{i \in B_k} \nabla f_i(x^k).$$

(c) What problem does mini-batch SGD solve?

Exercise 13.2

Implement

(i) the stochastic gradient descent method with I iterations, and $t_k = \frac{1}{\sqrt{k+1}}$,

(ii) The SAGA method with I iterations, and $t_k = \frac{1}{4}$,

Plot the iteration count vs. the empirical variance estimate for both methods. What does SAGA improve compared to SGD? What affect does this have on convergence rate compared to SGD?

Exercise 13.3

Let the following class of algorithms be

$$\begin{aligned} d^k &= \beta d^{k-1} + d(x^k, \xi_k), \qquad \xi_k \sim \xi \text{ independent}, \\ v_i^k &= \gamma v_i^{k-1} + \left[d(x^k, \xi_k)\right]_i^2, \qquad i = 1, \dots, n, \\ x_i^{k+1} &= x_i^k - t_k \frac{d_i^k}{\sqrt{v_i^k + \varepsilon}}, \qquad i = 1, \dots, n. \end{aligned}$$

For $x^0 \in \mathbb{R}^n$, $d^0 = v^0$, $\beta > 0$, $\gamma > 0$, $\epsilon = 10^{-9}$. Implement

(i) Adagrad with $\gamma = 1$, $\beta = 0.8$ and $t_k = t = \frac{1}{4} > 0$.

(ii) Adam with $\gamma = 0.9$, $\beta = 0.8$, $t_k = t(1-\beta)\sqrt{\frac{1-\gamma^k}{1-\gamma}}$, $t = \frac{1}{4}$.

- Plot the iteration count vs. the function value of the ergodic mean.
- Analyze what these two approaches modify compared to SGD.
- For a data science project of your choice, choose an algorithms of those discussed in this sheet and justify your choice.